

EXPERIMENT 11

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Menu-Driven Digit Operations Using switch-case

Problem: Input a positive integer and perform one operation according to the user's choice: sum of squares of digits using a **while** loop, sum of cubes using a **for** loop, or the difference between the sum of cubes and sum of squares using a **do-while** loop.

Algorithm

1. Start.
2. Input a positive integer **n**.
3. If **n** is not positive, display an error and stop.
4. Display the menu and input the user's choice.
5. Use **switch-case** to select the operation.
6. Choice 1: copy **n** to **temp**; use a **while** loop to extract each digit, add **digit x digit** to **sumSquares**, and remove the last digit.
7. Choice 2: use a **for** loop with **temp=n**; extract each digit and add **digit x digit x digit** to **sumCubes**.
8. Choice 3: initialize both sums to zero; use a **do-while** loop to calculate **sumSquares** and **sumCubes** together; compute **difference = sumCubes - sumSquares**.
9. Display the result. For an invalid menu choice, display an error message.
10. Stop.

Test Case Table

Test	Input n	Choice	Expected output
1	123	1	Sum of squares = 14
2	123	2	Sum of cubes = 36
3	123	3	Difference = 22
4	45	1	Sum of squares = 41
5	45	2	Sum of cubes = 189
6	7	3	Difference = 336
7	0	-	Error: enter a positive integer

Manual Verification of Given Example

For n = 123:

Squares: $1^2 + 2^2 + 3^2 = 1 + 4 + 9 = 14$

Cubes: $1^3 + 2^3 + 3^3 = 1 + 8 + 27 = 36$

Difference = $36 - 14 = 22$.

C Program

```
#include <stdio.h>

int main() {
    int n, choice, temp, digit;
    int sumSquares = 0, sumCubes = 0, difference;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    if (n <= 0) {
        printf("Error: enter a positive integer.\n");
        return 0;
    }

    printf("\nMENU\n");
    printf("1. Sum of squares of digits (while loop)\n");
    printf("2. Sum of cubes of digits (for loop)\n");
    printf("3. Difference: cubes - squares (do-while loop)\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice) {
        case 1:
            temp = n;
            while (temp > 0) {
                digit = temp % 10;
                sumSquares += digit * digit;
                temp /= 10;
            }
            printf("Sum of squares = %d\n", sumSquares);
            break;

        case 2:
            for (temp = n; temp > 0; temp /= 10) {
                digit = temp % 10;
                sumCubes += digit * digit * digit;
            }
            printf("Sum of cubes = %d\n", sumCubes);
            break;

        case 3:
            temp = n;
            do {
                digit = temp % 10;
                sumSquares += digit * digit;
                sumCubes += digit * digit * digit;
                temp /= 10;
            } while (temp > 0);

            difference = sumCubes - sumSquares;
            printf("Difference = %d\n", difference);
            break;

        default:
            printf("Invalid choice.\n");
    }

    return 0;
}
```

Compile and Execute

Save the program as **digit_operations.c**. In Linux/GCC, use:

```
gcc digit_operations.c -o digit_operations
./digit_operations
```

Execution and Testing

Run the compiled program separately for each prepared test case and compare the actual result with the expected result in the table.

Sample Run - Choice 1

```
Enter a positive integer: 123

MENU
1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference: cubes - squares (do-while loop)
Enter your choice: 1
Sum of squares = 14
```

Sample Run - Choice 2

```
Enter a positive integer: 123
Enter your choice: 2
Sum of cubes = 36
```

Sample Run - Choice 3

```
Enter a positive integer: 123
Enter your choice: 3
Difference = 22
```

Conclusion

The program correctly uses **switch-case** for menu selection and the specifically requested loop for each operation: while for squares, for for cubes, and do-while for the difference. The given example $n=123$ produces 14, 36, and 22 respectively.